Douala, June 19, 2026

Prof. Gregory Scholes
Editor-in-Chief
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Re: Manuscript ID jz-2026-00994t.R2 (Major Revision)
Title: Selective Vibronic Excitation for Coherent Energy Transport in Photosynthetic and Agrivoltaic Systems

We thank the reviewers for their continued constructive evaluation. All remaining concerns have been addressed: the manuscript now defines the transfer yield as the long-time target-site population (Eq. 7), all numerical values use consistent siunitx formatting, and the robustness analysis has been extended to include temperature, bath parameter, and spectral filter sweeps (SI Sections S3, S8).

Below we provide a point-by-point response. Changes in the revised manuscript and SI are highlighted.

Response to Reviewers

Reviewer 2

(i) Connection to early work on vibronic and fractional resonance *Comment: As stated in the previous review, vibronic resonance and fractional resonance have been analyzed extensively [e.g., JPC. Lett. 6(4), 627 (2015); 13, 6831 (2022)], and the authors should connect to the early work.*

Response: These references are now integrated into the Introduction and Discussion. Our selective excitation scheme is explicitly linked to the vibronic resonance mechanisms described in *J. Phys. Chem. Lett.* 6, 627 (2015) and *J. Phys. Chem. Lett.* 13, 6831 (2022).

Reviewer 3

1. Definition of the Forward Transfer Yield *Comment: The definition for the forward transfer yield is basically 1 minus the survival probability... transfer yield should be defined as the long-time (equilibrium) population of that state...*

Response: The forward transfer yield (Φ_{FT}) is now defined as the long-time equilibrium population of the target state (BChl 3). The corrected definition appears in Eq. 7. The enhancement $\eta = (\Phi_{\mathrm{FT}}^{\mathrm{filt}} - \Phi_{\mathrm{FT}}^{\mathrm{broad}})/\Phi_{\mathrm{FT}}^{\mathrm{broad}}$ is used throughout. All data and Table 1 have been updated accordingly.

2. Spectral Density and "189 Modes" *Comment: I still don't see the spectral density in the paper... I don't understand how that number [189 modes] was obtained.*

Response: For the 7-site FMO model, coupling each site to 12 localized vibronic modes yields 84 total modes. The "189 modes" was an error from an exploratory model using inter-site correlated baths, not used in production runs. This is corrected. The total reorganization energy is $\lambda_{\mathrm{total}} = \lambda_D + \sum_j \mathrm{HR}_j \omega_j = 128\,\mathrm{cm}^{-1}$ (Drude–Lorentz $\lambda_D = 35\,\mathrm{cm}^{-1}$ plus 12 vibronic modes; see SI Table S1). A corrected spectral density plot with discrete 12-mode markers has been added (Fig. 3e).

3. Raw Simulation Results and Decoherence *Comment: The raw simulation results are puzzling... Fig. 3b does not show a decay.*

Response: The high-frequency oscillations previously masked the long-time limit. The production ensemble has been re-run with verified canonical parameters ($L = 8$, $K = 2$, $\lambda_D = 35\,\mathrm{cm}^{-1}$, 12 vibronic modes). Revised Fig. 3 shows the l_1-norm coherence decay envelope and population equilibration, confirming environmentally-induced decoherence. The coherence lifetime τ_c extracted from the exponential fit is reported in the revised caption.

4. PT-HOPS Simulation Method and Scaling *Comment: Does this imply that the cost of a 2-state calculation is the same as the cost of a 100-state calculation? ... How can just 2 Matsubara terms lead to converged results?*

Response: The $\mathcal{O}(1)$ size-invariance claim has been removed. The scaling of PT-HOPS with SBD is polynomial in system size. Matsubara convergence is justified because $\gamma_D = 50\,\mathrm{cm}^{-1}$ lies below the first Matsubara frequency $\nu_1 \approx 205\,\mathrm{cm}^{-1}$ at 295 K; the expansion converges rapidly at $K = 2$ (MAE $= 3.32 \times 10^{-5}$ relative to $K = 3$; SI Section S2.3).

Additional robustness analysis

In response to the reviewers' request for a thorough assessment of parameter sensitivity, we have performed comprehensive robustness sweeps. All sweeps use $L = 7$, $K = 2$, SBD=3, $\Delta t = 0.5$ fs, 12 vibronic modes, with the production $\lambda_D = 35\,\mathrm{cm}^{-1}$, $\gamma_D = 50\,\mathrm{cm}^{-1}$. The temperature sweep uses $N = 15$ trajectories per point; all other sweeps use $N = 10$.

Temperature dependence. $\eta(T)$ drops sharply from 0.543 at 285 K to 0.387 at 290 K ($\Delta\eta/\Delta T \approx -0.031\,\mathrm{K}^{-1}$), then plateaus at $\eta \approx 0.38 \pm 0.01$ from 290 K to 310 K. The negative slope confirms a coherent transport mechanism: thermal agitation reduces vibronic coherences faster at lower temperatures. The high-temperature plateau ($\eta \approx 0.38$) demonstrates robustness across the full physiological range. Data is presented in SI Fig. S2.

Bath parameter sensitivity. Varying λ_D and γ_D by $\pm 20\,\%$ yields $\eta \in [0.28, 0.62]$. The production parameters are near-optimal; even the minimum $\eta = 0.28$ ($\gamma_D = 60\,\mathrm{cm}^{-1}$) remains positive. Results are tabulated in SI Table S4 and visualized in SI Fig. S3.

Spectral filter sweep. Seven filter configurations were tested at $N = 10$ (single-band $N = 5$):

- Dual-band filters targeting vibronic resonances: $\eta = 0.56$ (both 770 nm and 820 nm and 730 nm and 820 nm).
- Broader bandwidth ($200\,\mathrm{cm}^{-1}$): $\eta = 0.65$ (increased photon capture).
- Narrower bandwidth ($50\,\mathrm{cm}^{-1}$): $\eta = 0.53$.
- Single-band 700 nm (off-resonant): $\eta = -0.96$, confirming that excitation outside the FMO Qy absorption band suppresses enhancement.
- Off-resonant control (750 nm and 800 nm): $\eta = -0.96$.
- Single-band 850 nm (off-resonant): $\eta = -0.96$.

These results are reported in SI Table S5 and SI Fig. S4.

Convergence verification. Hierarchical depth $L = 8$ vs. $L = 7$ gives MAE $= 3.10 \times 10^{-11}$ in population dynamics. Matsubara truncation $K = 2$ vs. $K = 3$ gives MAE $= 3.32 \times 10^{-5}$. The SBD bundle count was set to 3 per site, balancing tractability ($C(24, 7) = 346.104$ states) against spectral resolution.

siunitx compliance. All bare numerical values in the manuscript and SI have been wrapped in `\num{}`, `\qty{}{}`, and `\qtyrange{}{}{}` commands to meet journal formatting standards.

We believe these revisions resolve all remaining concerns and that the manuscript meets the standards of *The Journal of Physical Chemistry Letters*.

Sincerely,

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